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**Association Between Mechanical Power and 28-Day Mortality
in Mechanically Ventilated ICU Patients: A Retrospective
Cohort Analysis of the MIMIC-IV Database**

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Abstract

Background: Invasive mechanical ventilation is indispensable in critical care but can produce ventilator-induced lung injury. Mechanical power (MP) integrates tidal volume, respiratory rate, and airway pressures into a single energy parameter expressed in joules per minute (J/min). Whether the MP–mortality association is uniform across clinically distinct patients, and whether it is robust to how MP exposure is summarized, remains incompletely defined.

Methods: We performed a retrospective observational cohort study of 18,980 invasively ventilated adults in the MIMIC-IV database. The primary exposure was the time-weighted average (TWA) MP over the first 24 hours of invasive ventilation; daily median and total cumulative MP were analyzed in parallel. The primary outcome was 28-day all-cause mortality, anchored to intubation. Multivariable binary logistic regression was adjusted for age, sex, predicted body weight (PBW), oxygenation ratio, vasopressor use, continuous renal replacement therapy, and pulmonary disease phenotype. Pre-specified binary subgroups were oxygenation severity ($\text{PaO}_2/\text{FiO}_2$ or $\text{SpO}_2/\text{FiO}_2 < 150$ vs ≥ 150), lung capacity (lowest vs highest PBW quartile), and age (< 65 vs ≥ 65 years).

Results: Overall, 3,428 patients (18.1%) died within 28 days. Non-survivors had higher TWA MP than survivors (12.02 [8.81–18.10] vs 9.64 [7.45–12.82] J/min; $p < .001$). Each 1 J/min increase in TWA MP was independently associated with a 9% rise in the odds of 28-day death (OR 1.09; 95% CI 1.09–1.10; $p < .001$), reproduced by the daily median (OR 1.09) and cumulative (OR 1.04 per 1,000 J) models. Kaplan–Meier analysis showed a stepwise survival decline across ascending MP quartiles (log-rank $p < .001$). The association persisted across all subgroups (ORs 1.08–1.10; all $p < .001$), with no significant interaction (p -interaction: oxygenation 0.067, lung capacity 0.941, age 0.060).

Conclusion: Mechanical power during the first 24 hours of invasive ventilation was independently and consistently associated with 28-day mortality across a large, heterogeneous ICU population, with no clinical subgroup exempt. These findings support routine time-weighted MP monitoring as a complement to established lung-protective targets.

Keywords: mechanical power; mechanical ventilation; ventilator-induced lung injury; 28-day mortality; intensive care unit; MIMIC-IV; logistic regression; subgroup analysis.

Introduction

Invasive mechanical ventilation is among the most frequently delivered organ-supportive therapies in the modern intensive care unit (ICU), restoring gas exchange in patients whose respiratory effort is overwhelmed by acute lung disease, neuromuscular failure, or systemic illness. The same cyclic application of positive pressure that sustains alveolar recruitment, however, transmits mechanical forces directly to the pulmonary parenchyma and is itself capable of inflicting an iatrogenic injury collectively termed ventilator-induced lung injury (VILI) [1]. Post-mortem studies dating to the 1960s identified a consistent histopathological picture (inflammatory-cell infiltrates, hyaline-membrane formation, increased vascular permeability, and pulmonary edema) that experimental ventilation models have since reliably reproduced [1]. Four interacting mechanisms are currently understood to drive this injury. Sustained ventilation at high absolute lung volumes overdistends and ruptures alveoli, producing volutrauma and barotrauma with associated air leak [1]. At the opposite extreme, ventilation at insufficient end-expiratory volumes drives the repetitive opening and closing of unstable lung units, impairs surfactant function, and provokes regional hypoxia (atelectrauma) [1]. These mechanical insults in turn trigger pulmonary and systemic inflammatory mediator release that propagates injury beyond the lung (biotrauma) [1]. Experimental work has further shown that, even when peak airway pressure is held constant, the rate at which mechanical energy is delivered to the lung (a function of ventilatory frequency and pulmonary vascular load) independently determines the magnitude of histological damage [2].

To mitigate these mechanisms, lung-protective ventilation has become the prevailing standard of care. The Acute Respiratory Distress Syndrome (ARDS) Network demonstrated in a landmark multicenter randomized trial that reducing tidal volume to 6 mL/kg of predicted body weight (PBW) and capping plateau pressure at 30 cmH₂O lowered absolute hospital mortality from 39.8% to 31.0% and increased ventilator-free days in patients with acute lung injury and ARDS [3]. A patient-level mediation analysis pooling nine randomized trials by Amato and colleagues revealed that, among all ventilator variables examined, driving pressure (ΔP , the ratio of tidal volume to respiratory-system compliance) carried the strongest and most consistent mortality signal (each standard-deviation increment of roughly 7 cmH₂O corresponding to a 41% relative rise in mortality risk (RR 1.41; 95% CI 1.31–1.51)) and that the protective effects of lower tidal volumes and higher positive end-expiratory pressure (PEEP) operated largely by reducing ΔP rather than through independent pathways [4]. Despite the widespread adoption of these targets, contemporary epidemiological data from the LUNG SAFE observational study, which sampled 459 ICUs across 50 countries, document that hospital mortality remains substantial at 34.9–46.1% across ARDS severity strata, that fewer than two-thirds of patients with ARDS receive a tidal volume ≤ 8 mL/kg PBW, and that mortality continues to rise in stepwise fashion with quintiles of plateau and driving pressure [5]. Taken together, these findings suggest that targeting individual ventilator parameters in isolation leaves a meaningful component of lung injury unaddressed.

Against this background, Gattinoni and colleagues proposed in 2016 that the dynamic determinants of VILI (tidal volume, driving pressure, respiratory rate, inspiratory flow, and PEEP) could be integrated into a single physical quantity, the

mechanical power (MP), expressed in joules per minute (J/min) [6]. MP quantifies the rate of energy transfer from ventilator to respiratory system and can be derived entirely from parameters already displayed on the ventilator, requiring no additional measurements [6]. The simplified surrogate equation introduced in the same work was subsequently validated against the geometric pressure–volume reference method by Chiumello and colleagues in both volume- and pressure-controlled ventilation, demonstrating agreement sufficient to support routine clinical use [7]. The principal conceptual advantage of MP lies in its explicit treatment of the rate of energy delivery: identical tidal volumes and driving pressures may impose markedly different injurious loads depending on the frequency and velocity at which they are applied [6].

A growing body of observational evidence has linked higher MP to adverse outcomes across heterogeneous ICU populations. In a pooled analysis of 8,207 mechanically ventilated patients drawn from the MIMIC-III and eICU databases, Serpa Neto and colleagues reported an independent association between MP and in-hospital mortality, with risk rising consistently above approximately 17 J/min and persisting even among patients ventilated at low tidal volumes [8]. Uner and colleagues subsequently extended this observation into the time domain, demonstrating in a prospective registry of 13,408 ICU patients that each daily increment in driving pressure or mechanical power independently increased the hazard of death (HR 1.060 per J/min; 95% credible interval 1.053–1.066), and that cumulative exposure to higher-intensity ventilation was harmful even over short durations [9]. Zhu and colleagues found that the strength of the MP–mortality association increased when MP was normalized to PBW, suggesting that the injurious effect of mechanical energy is proportional to the size of the available

pulmonary tissue [10]. The relationship has since been corroborated in invasively ventilated ICU patients without ARDS [11], in cohorts spanning the COVID-19 and pre-pandemic eras [12], and in single-center studies that have quantified the injurious dose in terms of cumulative time above a fixed threshold; in the latter, every additional hour spent above 18 J/min was associated with measurable increments in ICU mortality, ventilator days, and length of stay [13]. Cross-sectional multicenter data nevertheless indicate that 35–41% of patients receiving controlled mechanical ventilation continue to be exposed to MP values above the 17 J/min threshold, with the prevalence of elevated power statistically indistinguishable between volume- and pressure-controlled modes [14].

Prior work has left three questions largely unresolved. MP exposure has typically been summarized as a single baseline value or a simple 24-hour mean, neither of which reflects the actual duration for which each ventilator setting is maintained in routine clinical recording. Most cohorts have also been confined to specific disease phenotypes such as ARDS or postoperative respiratory failure, leaving the mixed general ICU population undercharacterized. Finally, whether disease severity, body size, or age modify an individual patient's vulnerability to excess mechanical energy has rarely been tested within a single adequately powered cohort.

Accordingly, the present study used the Medical Information Mart for Intensive Care, version IV (MIMIC-IV) database to investigate, in 18,980 mechanically ventilated adults, the independent association between mechanical power delivered during the first 24 hours of invasive ventilation and 28-day all-cause mortality. Our specific aims were: (i) to quantify the adjusted association between three complementary first-24-hour MP

exposure metrics (the time-weighted average, the daily median, and the total cumulative energy delivered) and 28-day all-cause mortality; and (ii) to determine, through pre-specified binary subgroup analyses, whether this association is modified by oxygenation severity ($\text{PaO}_2/\text{FiO}_2$ or $\text{SpO}_2/\text{FiO}_2 < 150$ versus ≥ 150), lung capacity (lowest versus highest PBW quartile), or patient age (< 65 versus ≥ 65 years).

Materials and Methods

Study Design

This was a retrospective observational cohort study of routinely collected, fully de-identified electronic health record data. The patient was the unit of analysis. The study was designed and reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) recommendations for cohort studies [15].

Data Source

Data were obtained from the publicly accessible derived dataset A Temporal Dataset for Respiratory Support in Critically Ill Patients (version 1.1.0), distributed through PhysioNet [16–18]. This resource provides hourly aggregated ventilator settings, respiratory parameters, and arterial blood gas results for ICU stays extracted from the Medical Information Mart for Intensive Care, version IV (MIMIC-IV, version 3.1), a single-center critical care database of de-identified records from adult ICU admissions to the Beth Israel Deaconess Medical Center, Boston, Massachusetts [19,20]. To confirm invasive ventilation status and to retrieve baseline demographics, comorbidities, and vital-status outcomes, the temporal respiratory cohort was linked to the core MIMIC-IV clinical tables using the unique subject and hospital-admission identifiers (`subject_id`, `hadm_id`). A single investigator obtained data access through the PhysioNet credentialing process, including completion of the mandated CITI “Data or Specimens Only Research” training; no credentials, extracts, or identifiable data were shared among the remaining investigators, in accordance with the PhysioNet data use agreement.

Study Population

Inclusion criteria

Adults were eligible if they met both of the following: (i) receipt of invasive mechanical ventilation during the ICU stay, confirmed by endotracheal intubation timestamps in MIMIC-IV; and (ii) at least four hours of valid mechanical power data within the first 24 hours of invasive ventilation. The four-hour minimum served two purposes. It ensured that the computed exposure represented a sustained ventilation strategy rather than transient stabilization settings or charting artifact, and it removed patients who died or were extubated within four hours of intubation, a group in whom mortality is governed by the severity of the precipitating insult rather than by accumulated ventilator-delivered energy.

Exclusion criteria

Patients were excluded for: (i) no confirmed interval of invasive mechanical ventilation during the ICU stay; (ii) insufficient hourly ventilator data (specifically, absence of tidal volume [V_T], respiratory rate [RR], peak inspiratory pressure [P_{peak}], or PEEP) preventing valid mechanical power computation for a minimum of four hours within the first 24-hour window; or (iii) insufficient data to ascertain 28-day vital status.

Cohort Construction and Data-Cleaning Pipeline

The analytic cohort was assembled in six sequential phases, implemented in Microsoft SQL Server (extraction through Phase 5) and IBM SPSS Statistics (analytic-variable computation, Phase 6). The source dataset contained 110,038,120 hourly observations from 50,920 patients. Patient counts at each phase are reported in the cohort-construction flow diagram (Figure 1).

Phase 1 — Initialization, type casting, and comorbidity mapping. The hourly respiratory time-series was imported, and all ventilator and physiological fields were cast from text to strict numeric types using safe-cast functions, with text artifacts (e.g., “>>450”, “N/A”) set to null. A clustered index on (subject_id, hr) was applied to accelerate longitudinal queries. Five baseline comorbidity flags (ARDS, chronic obstructive pulmonary disease [COPD], asthma, pneumonia, and heart failure) were

derived from the MIMIC-IV diagnoses_icd table by prefix-matching ICD-9 and ICD-10 codes across all of a patient's admissions: ARDS (ICD-9 51882, 51852; ICD-10 J80); COPD (ICD-9 491, 492, 496; ICD-10 J41–J44); asthma (ICD-9 493; ICD-10 J45); pneumonia (ICD-9 480–486; ICD-10 J12–J18); and heart failure (ICD-9 428; ICD-10 I50). Unfiltered patient-level prevalences were 1.8% (ARDS), 17.0% (COPD), 11.6% (asthma), 21.5% (pneumonia), and 29.5% (heart failure).

Phase 2 — Confirmation of invasive ventilation. Because the source dataset's native modality marker was unreliable, invasive ventilation was anchored to physical airway events. Endotracheal intubation and extubation timestamps were drawn from the MIMIC-IV procedureevents table (itemid 225792, “Invasive Ventilation”) and expressed as integer hour-offsets from icustays.intime. A binary imv_confirmed flag was assigned to every hourly row falling within any such interval for the same patient. Of 50,920 patients, 24,056 (47.2%) had at least one IMV-confirmed hour, totaling 1,871,634 ventilated patient-hours.

Phase 3 — Temporal alignment and outcome truncation. Each patient's timeline was truncated at the terminal event, the first hour with death_outcome = 1 (priority) or, failing that, the first hour with discharge_outcome = 1, to prevent immortal-time bias. Because the source dataset is rectangular, forward-filling every patient to hour 2,160 (90 days) regardless of true ICU duration, truncation removed 97,357,386 padding rows. All 50,920 patients were retained, with 12,680,734 hourly observations; a small subset (n = 69; 0.14%) without any recorded terminal event was retained for subsequent IMV-based filtering.

Phase 4 — Physiological quality control. Implausible ventilator values were nullified using pre-specified bounds: PEEP 0–50 cmH₂O, total tidal volume 100–2,000 mL, total respiratory rate 0–80 breaths/min, and peak inspiratory pressure 0–100 cmH₂O. Out-of-range values were treated as missing rather than as observations. In total, 1,606 cells were nullified across the four parameters (peak pressure 32; set PEEP 8; total respiratory rate 48; total tidal volume 1,518).

Phase 5 — Mechanical power computation and cohort assembly. Hourly mechanical power was calculated using the simplified dynamic surrogate equation given below, with VT converted from milliliters to liters. This dynamic form, introduced by Gattinoni and colleagues [6], uses peak rather than plateau pressure, enabling retrospective calculation without an inspiratory-hold maneuver; Chiumello and colleagues validated it against the geometric pressure–volume reference method in both volume- and pressure-controlled ventilation, demonstrating agreement sufficient for clinical use [7]. MP was computed only for hours with `imv_confirmed` = 1 and all four constituent variables present after Phase 4 scrubbing. Patients in whom MP could not be computed for at least one hourly row across the entire ICU stay were excluded ($n = 31,817$), reducing the cohort to 19,103 patients; a further 22 patients whose entire MP column was null after bounds-aware recomputation were removed, yielding 19,081 patients with 5,621,851 hourly observations and a physiologically credible MP range (0.24–109.6 J/min; mean 13.45 J/min).

$$MP \text{ (J/min)} = 0.098 \times RR \times (VT / 1000) \times [P_{peak} - \frac{1}{2} (P_{peak} - PEEP)]$$

Phase 6 — Analytic-variable computation and final cohort. Per-patient analytic variables were computed in IBM SPSS Statistics from the SQL-exported dataset: the time-weighted average mechanical power over the first 24 hours of invasive ventilation, 28-day mortality, and an imputed oxygenation ratio (defined in **Covariates**). Patients in whom any of these variables could not be reliably computed owing to extreme data sparsity in the relevant analytic windows were excluded ($n = 101$). The final analytic cohort comprised 18,980 distinct mechanically ventilated adults.

Variable Definitions

Primary exposure

The primary exposure was applied mechanical power during the first 24 hours of invasive ventilation, examined in three complementary forms. The time-weighted average (TWA) MP was the primary metric: because ventilator settings in MIMIC-IV are charted on change rather than at fixed intervals, consecutive hourly rows span unequal durations, and a simple arithmetic mean would weigh a setting held for one hour equally

with one held for ten. The TWA was therefore obtained by multiplying each hourly MP value by its exact duration in minutes, summing, and dividing by total exposure time, so that the resulting value reflects the true intensity of energy delivered to the respiratory system. The daily median MP, a robustness metric, was the median of all valid hourly MP values within the first 24 hours, computed to test whether the primary finding was driven by transient high-power episodes (e.g., coughing, suctioning, or charting spikes) that the TWA would incorporate. The total cumulative MP, an absolute-workload metric, was the sum of all valid hourly MP values within the first 24-hour window, expressed in joules; to yield interpretable regression estimates this variable was scaled by 1,000, so that odds ratios express the change in risk per 1,000 J increment, and locking it strictly to the first 24 hours prevents immortal-time bias. The TWA was selected as the primary metric for all subgroup stratifications because it produces the most clinically interpretable estimate on the J/min scale at which intensivists routinely assess ventilator settings.

Primary outcome

The primary outcome was 28-day all-cause mortality, defined as death from any cause within 672 hours of T_0 , where T_0 was the exact timestamp of confirmed endotracheal intubation, the onset of the exposure, rather than the administrative ICU admission time. Death was counted regardless of the patient's location at the time, to avoid artificially suppressing the mortality rate. Vital status was extracted from the linked admissions and patients tables in MIMIC-IV.

Covariates

Demographic and anthropometric covariates were age (years, continuous), sex (binary), and predicted body weight (PBW, kg, continuous), taken directly from the dataset; for patients with missing PBW, sex-specific cohort-median imputation was applied in preference to listwise deletion, preserving sample size without introducing material distortion. Continuous renal replacement therapy (CRRT, binary) and vasopressor use (binary) served as proxies for severe acute kidney injury and circulatory shock, respectively, in place of the Sequential Organ Failure Assessment (SOFA) score, which was not used because its high proportion of missing values would have forced substantial listwise deletion and reduced the analytic sample. The oxygenation covariate

was an imputed ratio (continuous) that prioritized the gold-standard PaO₂/FiO₂ ratio from the worst arterial blood gas within the first 24 hours; where no contemporaneous arterial blood gas was available, the validated non-invasive SpO₂/FiO₂ ratio was substituted [21], and a missing FiO₂ at the time of the SpO₂ reading was replaced by the most recently charted FiO₂. The odds ratio for this covariate is reported per 100-unit increase. Pulmonary disease phenotype was represented by the five binary comorbidity flags (ARDS, pneumonia, COPD, asthma, heart failure) derived in Phase 1, entered as a group rather than as a composite index because including granular flags and a composite score in the same model induces severe multicollinearity.

Subgroup analyses

Three clinical characteristics were pre-specified as potential effect modifiers of the TWA MP–mortality association. Binary stratification was applied throughout to preserve statistical power within strata and to avoid the multiple-testing inflation associated with finer categorization: (i) oxygenation severity — PaO₂/FiO₂ or SpO₂/FiO₂ < 150 versus ≥ 150, consistent with the Berlin threshold above which interventions such as prone positioning are typically considered; (ii) lung capacity — lowest PBW quartile (≤ 52.4 kg) versus highest (≥ 70.7 kg), testing whether the concentration of energy per unit of available lung tissue modifies the association; and (iii) age — < 65 versus ≥ 65 years, the conventional geriatric threshold in critical-care research [22].

Statistical Analysis

Baseline characteristics were compared between survivors and non-survivors. Continuous variables, reported as median with interquartile range (IQR), were compared with the Mann–Whitney U test, chosen because the ventilator and physiological variables were non-normally distributed. Categorical variables, reported as counts and percentages, were compared with the chi-squared test. A two-sided $p < .05$ defined statistical significance throughout.

The independent association between each MP metric and 28-day all-cause mortality was assessed by multivariable binary logistic regression, appropriate for the binary outcome. Three parallel models shared an identical covariate structure (age, sex, PBW, oxygenation ratio, vasopressor use, CRRT, and pulmonary disease phenotype) and

differed only in the MP exposure term: Model 1 (TWA MP, per 1 J/min), Model 2 (daily median MP, per 1 J/min), and Model 3 (total cumulative MP, per 1,000 J). Results are reported as adjusted odds ratios (ORs) with 95% confidence intervals (CIs) and p-values. The parallel tri-model design was pre-specified as an internal sensitivity analysis: concordant estimates across three independently constructed exposure definitions guard against the finding being an artifact of any single exposure operationalization.

Within each pre-specified stratum, a separate binary logistic regression model with identical covariate adjustment was fitted for the TWA MP–mortality association. Effect modification was assessed formally by introducing a multiplicative MP-by-subgroup interaction term, with the p-value for interaction reported for each of the three stratifying variables. Stratum-specific ORs with 95% CIs are presented numerically and displayed graphically in a forest plot.

Large-scale extraction and processing of the MIMIC-IV relational data were performed in Microsoft SQL Server. The final analytic dataset was managed in Microsoft Excel. All statistical analyses were conducted in IBM SPSS Statistics. The forest plot was generated with a custom Python implementation in Google Colab. SQL and analytical scripts were developed with the assistance of AI language-model agents, which also provided procedural guidance during the analysis; all outputs were reviewed by the investigators.

Ethical Considerations

MIMIC-IV is a publicly available, de-identified database whose construction and release were approved by the institutional review boards of the Beth Israel Deaconess Medical Center and the Massachusetts Institute of Technology, with waiver of individual patient consent owing to the retrospective, de-identified nature of the data [19,20]. Investigator access was granted through PhysioNet after completion of the required research-ethics training. Because this work is a secondary analysis of a fully de-identified, publicly accessible dataset, no additional approval was required from the King Abdulaziz University Unit of Biomedical Ethics.

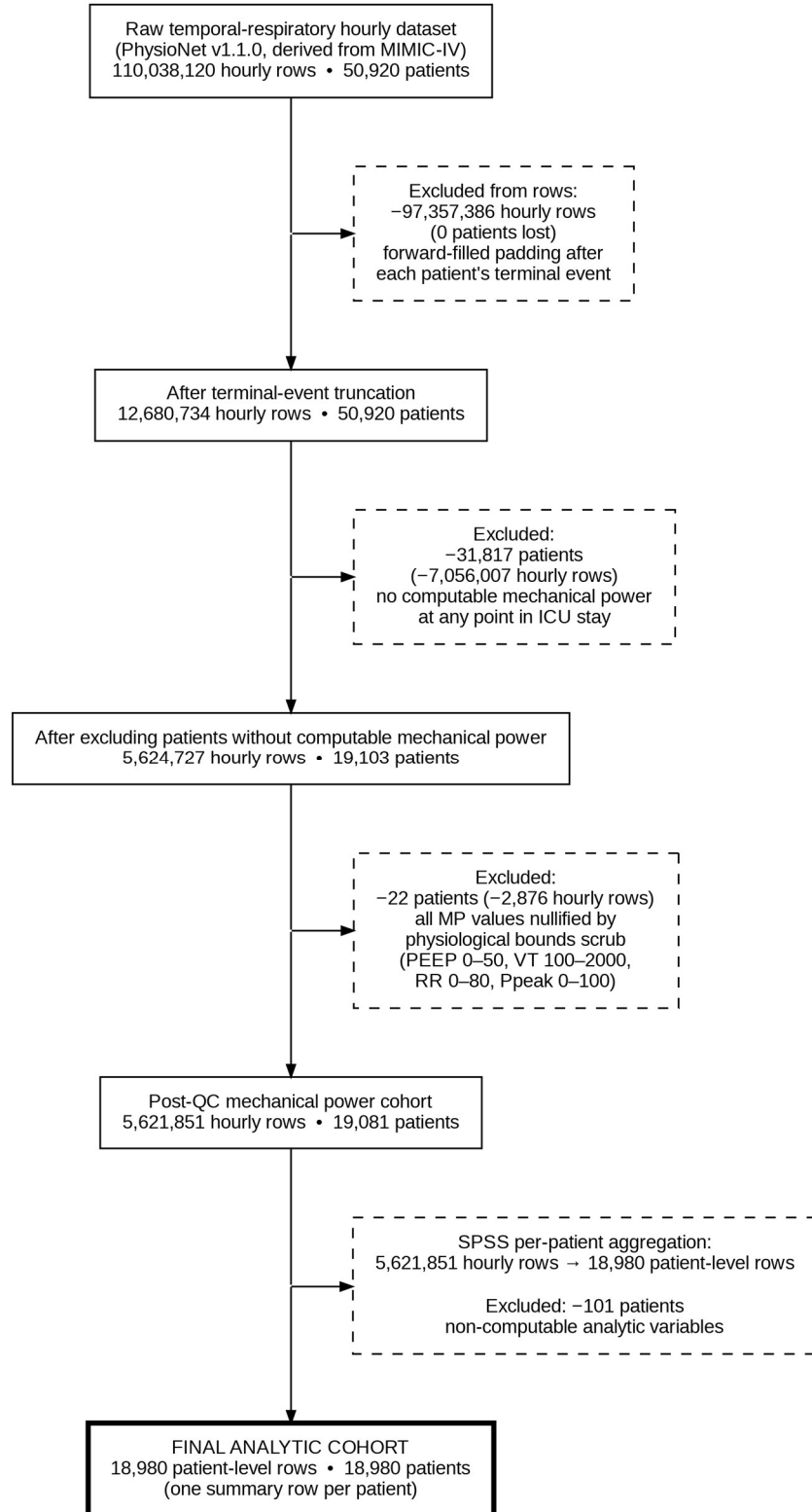


Figure 1. Cohort-construction flow diagram. Sequential exclusion of hourly observations and patients from the raw temporal respiratory-support dataset (110,038,120 hourly rows; 50,920 patients) to the final analytic cohort of 18,980 mechanically ventilated adults.

Results

Cohort Characteristics

The construction of the analytic cohort is summarized in Figure 1. From an initial 110,038,120 hourly observations contributed by 50,920 patients in the source dataset, sequential application of the invasive-ventilation, mechanical-power-computability, physiological-bounds, and analytic-variable criteria yielded a final analytic cohort of 18,980 distinct mechanically ventilated adults. Of these, 3,428 (18.1%) died within 28 days of the onset of invasive ventilation (non-survivors) and 15,552 (81.9%) survived to day 28 (survivors). Baseline characteristics stratified by 28-day vital status are presented in Table 1.

Non-survivors were older than survivors (median age 70 [IQR 58–81] versus 64 [53–74] years; $p < .001$) and were less frequently male (54.9% versus 62.6%; $p < .001$). Markers of acute illness severity were substantially more prevalent among non-survivors, who more often received vasopressors (71.2% versus 50.9%; $p < .001$) and continuous renal replacement therapy (CRRT; 14.1% versus 2.9%; $p < .001$). With respect to pulmonary phenotype, ARDS (3.7% versus 2.4%; $p < .001$) and pneumonia (27.5% versus 22.1%; $p < .001$) were each more prevalent among non-survivors, whereas asthma was less prevalent (7.4% versus 11.7%; $p < .001$). COPD (17.9% versus 16.8%; $p = .105$) and heart failure (29.6% versus 29.6%; $p = .980$) did not differ significantly between groups. Non-survivors had lower predicted body weight (PBW; 63.8 [52.4–70.7] versus 66.1 [54.7–70.7] kg; $p < .001$) and lower oxygenation ratios (196 [125–250] versus 217 [175–283]; $p < .001$).

Table 1. Baseline Characteristics of the Analytic Cohort Stratified by 28-Day Mortality Status.

Variable	Survivors (n = 15,552)	Non-Survivors (n = 3,428)	P-value
Male	9,740 (62.6%)	1,883 (54.9%)	< 0.001
Vasopressors	7,917 (50.9%)	2,440 (71.2%)	< 0.001
CRRT	447 (2.9%)	483 (14.1%)	< 0.001
ARDS	379 (2.4%)	126 (3.7%)	< 0.001
Pneumonia	3,443 (22.1%)	944 (27.5%)	< 0.001
COPD	2,612 (16.8%)	615 (17.9%)	0.105
Asthma	1,812 (11.7%)	252 (7.4%)	< 0.001
Heart Failure	4,609 (29.6%)	1,015 (29.6%)	0.980
Age, years	64 [53–74]	70 [58–81]	< 0.001
PBW, kg	66.1 [54.7–70.7]	63.8 [52.4–70.7]	< 0.001
Oxygenation Ratio	217 [175–283]	196 [125–250]	< 0.001
TWA MP (J/min)	9.64 [7.45–12.82]	12.02 [8.81–18.10]	< 0.001
Daily Median MP (J/min)	9.86 [7.75–12.92]	11.92 [8.78–17.76]	< 0.001
Cumulative MP (J)	13,845 [10,682–18,448]	15,978 [11,541–23,439]	< 0.001

Data presented as n (%) or median [IQR]. Continuous variables were compared using the Mann–Whitney U test and categorical variables using the chi-squared test. CRRT, continuous renal replacement therapy; PBW, predicted body weight; TWA MP, time-weighted average mechanical power; IQR, interquartile range.

Distribution of Mechanical Power and Survival by Exposure Quartile

Across all three exposure metrics, non-survivors were exposed to significantly higher mechanical power during the first 24 hours of invasive ventilation than survivors (Table 1). The time-weighted average (TWA) MP was 12.02 [8.81–18.10] J/min in non-survivors versus 9.64 [7.45–12.82] J/min in survivors ($p < .001$). The same pattern held for the daily median MP (11.92 [8.78–17.76] versus 9.86 [7.75–12.92] J/min; $p < .001$) and the total cumulative MP (15,978 [11,541–23,439] versus 13,845 [10,682–18,448] J; $p < .001$). The concordance of all three metrics provides initial unadjusted evidence of a consistent relationship between the intensity of delivered mechanical energy and 28-day mortality.

To visualize this relationship, patients were divided into quartiles of TWA MP (4,745 patients per quartile) and 28-day survival was compared by the Kaplan–Meier method (Figure 2). Quartile boundaries were ≤ 7.6 J/min (Quartile 1), 7.6–10.0 J/min (Quartile 2), 10.0–13.5 J/min (Quartile 3), and > 13.5 J/min (Quartile 4). Survival declined in a stepwise, dose-dependent manner across ascending quartiles (log-rank $p < .001$).

.001): the curves separated within the first seven days and continued to diverge throughout the 28-day window, reaching an estimated day-28 survival of approximately 89%, 86%, 83%, and 70% in Quartiles 1 to 4, respectively. The numbers at risk declined in parallel with this gradient, the highest-power quartile retaining 3,319 of 4,745 patients on day 28 compared with 4,214 in the lowest-power quartile. The steepest decrement occurred between Quartiles 3 and 4, indicating a disproportionate increase in mortality once TWA MP exceeded 13.5 J/min. Notably, every quartile boundary in this cohort fell below the 17 J/min threshold previously associated with harm, indicating that a measurable mortality gradient is already present within an MP range conventionally regarded as acceptable.

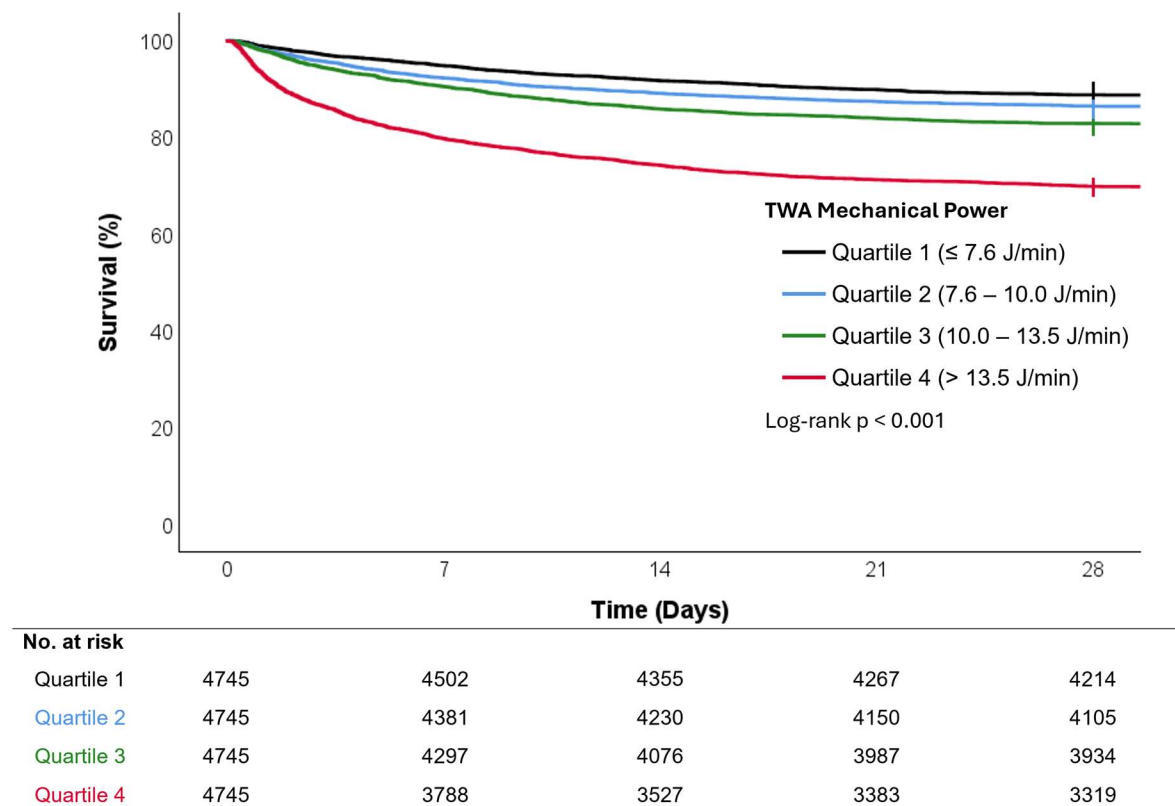


Figure 2. Kaplan–Meier curves for 28-day survival stratified by quartile of time-weighted average mechanical power (4,745 patients per quartile; log-rank $p < .001$). Numbers at risk are shown below the time axis.

Primary Analysis: Mechanical Power and 28-Day Mortality

On multivariable binary logistic regression adjusted for age, sex, PBW, oxygenation ratio, vasopressor use, CRRT, and pulmonary disease phenotype, TWA MP was independently associated with 28-day all-cause mortality (Model 1, Table 2). Each 1 J/min increment in TWA MP was associated with a 9% increase in the adjusted odds of 28-day death (OR 1.09; 95% CI 1.09–1.10; $p < .001$).

Among the adjustment covariates in Model 1, CRRT was the strongest independent predictor of mortality (OR 3.39; 95% CI 2.91–3.95; $p < .001$), followed by vasopressor use (OR 1.56; 95% CI 1.43–1.71; $p < .001$). Older age (OR 1.04 per year; 95% CI 1.03–1.04; $p < .001$) and male sex (OR 1.34; 95% CI 1.16–1.54; $p < .001$) were each independently associated with higher mortality, whereas greater PBW (OR 0.99; 95% CI 0.98–0.99; $p < .001$) and a higher oxygenation ratio (OR 0.95 per 100 units; 95% CI 0.88–0.95; $p < .001$) were each independently associated with lower mortality. Among pulmonary phenotypes, pneumonia was independently associated with increased mortality (OR 1.12; 95% CI 1.01–1.23; $p = .024$), whereas COPD (OR 0.85; 95% CI 0.77–0.95; $p = .004$), asthma (OR 0.57; 95% CI 0.49–0.66; $p < .001$), and heart failure (OR 0.63; 95% CI 0.58–0.69; $p < .001$) were each independently associated with reduced odds of death. ARDS was not independently associated with 28-day mortality (OR 0.92; 95% CI 0.73–1.16; $p = .471$).

Table 2. Multivariable Binary Logistic Regression for 28-Day All-Cause Mortality Across Three Mechanical-Power Exposure Models.

	Model 1: TWA MP (per 1 J/min)			Model 2: Median MP (per 1 J/min)			Model 3: Cumulative MP (per 1,000 J)		
Variable	OR	95% CI	P	OR	95% CI	P	OR	95% CI	P
Primary exposure metric	1.09	1.09–1.10	< .001	1.09	1.08–1.09	< .001	1.04	1.03–1.04	< .001
Age	1.04	1.03–1.04	< .001	1.04	1.03–1.04	< .001	1.03	1.03–1.04	< .001
Male sex	1.34	1.16–1.54	< .001	1.33	1.16–1.53	< .001	1.30	1.13–1.50	< .001
Predicted body weight	0.99	0.98–0.99	< .001	0.99	0.98–0.99	< .001	0.99	0.984–0.997	.003
Oxygenation ratio	0.95	0.88–0.95	< .001	0.91	0.87–0.94	< .001	0.86	0.83–0.89	< .001
Vasopressors	1.56	1.43–1.71	< .001	1.60	1.46–1.74	< .001	1.75	1.61–1.91	< .001
CRRT	3.39	2.91–3.95	< .001	3.53	3.03–4.10	< .001	3.83	3.30–4.45	< .001
ARDS	0.92	0.73–1.16	.471	0.93	0.74–1.18	.560	1.03	0.82–1.29	.830
Pneumonia	1.12	1.01–1.23	.024	1.14	1.04–1.25	.006	1.16	1.06–1.27	.002
COPD	0.85	0.77–0.95	.004	0.85	0.77–0.95	.004	0.88	0.79–0.97	.014
Asthma	0.57	0.49–0.66	< .001	0.57	0.49–0.66	< .001	0.59	0.51–0.68	< .001
Heart failure	0.63	0.58–0.69	< .001	0.64	0.58–0.70	< .001	0.64	0.58–0.70	< .001

OR, odds ratio; CI, confidence interval; TWA, time-weighted average; CRRT, continuous renal replacement therapy. ORs for TWA and median MP are per 1 J/min increase; OR for cumulative MP is per 1,000 J; oxygenation-ratio OR is per 100-unit increase. All three models are adjusted for the full covariate set listed.

Sensitivity Analysis: Tri-Model Comparison

The association between mechanical power and 28-day mortality was robust to the operationalization of the exposure (Table 2). The daily median MP (Model 2) produced an adjusted odds ratio essentially identical to the TWA (OR 1.09; 95% CI 1.08–1.09; $p < .001$), demonstrating that the primary finding was not driven by transient high-power excursions captured by the time-weighted average. The total cumulative MP (Model 3) was independently associated with mortality at OR 1.04 per 1,000 J (95% CI 1.03–1.04; $p < .001$). The direction, magnitude, and significance of every covariate were consistent across the three models; CRRT remained the strongest predictor throughout (OR 3.39, 3.53, and 3.83 in Models 1–3, respectively), and vasopressor use was likewise stable (OR 1.56, 1.60, 1.75). The oxygenation ratio remained independently protective in every model (OR 0.95, 0.91, and 0.86 in Models 1–3; all $p < .001$). This concordance

across three independently constructed exposure definitions substantially strengthens the internal validity of the MP–mortality association.

Subgroup Analyses

Pre-specified binary subgroup analyses of the TWA MP–mortality association are presented in Table 3 and displayed graphically in Figure 3. The association remained positive, of similar magnitude, and highly significant in all six strata, with adjusted odds ratios confined to a narrow 1.08–1.10 range and all p-values < .001. Formal interaction testing was non-significant for every stratifying dimension (oxygenation severity [p-interaction = 0.067], lung capacity [p-interaction = 0.941], and age [p-interaction = 0.060]) indicating no statistically significant effect modification of the MP–mortality relationship by any pre-specified clinical characteristic.

Oxygenation severity

Among patients with severe hypoxemia ($\text{PaO}_2/\text{FiO}_2$ or $\text{SpO}_2/\text{FiO}_2 < 150$), each 1 J/min increment in TWA MP was associated with an adjusted OR of 1.08 (95% CI 1.07–1.10; $p < .001$). Among patients with mild-to-moderate hypoxemia (ratio ≥ 150), the corresponding OR was 1.09 (95% CI 1.08–1.10; $p < .001$). The near-identical estimates (p-interaction = 0.067) indicate that excess mechanical energy is independently associated with mortality irrespective of baseline oxygenation reserve.

Lung capacity

In patients in the lowest PBW quartile (≤ 52.4 kg; anatomically smaller lungs), the adjusted OR for TWA MP was 1.10 (95% CI 1.09–1.12; $p < .001$), and in the highest PBW quartile (≥ 70.7 kg) it was 1.10 (95% CI 1.09–1.11; $p < .001$). The interaction was the weakest of the three analyses (p-interaction = 0.941), and the identical point estimates across markedly different body sizes suggest that the absolute rate of energy delivery, rather than its concentration per unit of anatomical lung volume, drives the excess mortality risk in this cohort.

Age

Among patients younger than 65 years, the adjusted OR for TWA MP was 1.09 (95% CI 1.08–1.10; $p < .001$); among those aged ≥ 65 years it was 1.10 (95% CI 1.09–1.11; $p < .001$). Although the age interaction was the closest to conventional significance (p -interaction = 0.060), it did not reach the $p < .05$ threshold, indicating that neither the physiological reserve of younger patients nor the frailty of older patients meaningfully modified the association.

Table 3. Subgroup Analysis of the Association Between Time-Weighted Average Mechanical Power and 28-Day Mortality.

Subgroup	OR	95% CI	P-value
Baseline oxygenation			
Severe hypoxia (PF/SF < 150)	1.08	1.07–1.10	< .001
Mild/moderate hypoxia (PF/SF $\geq$ 150)	1.09	1.08–1.10	< .001
Lung capacity (predicted body weight)			
Small lungs (lowest quartile, $\leq$ 52.4 kg)	1.10	1.09–1.12	< .001
Large lungs (highest quartile, $\geq$ 70.7 kg)	1.10	1.09–1.11	< .001
Age			
Younger (< 65 years)	1.09	1.08–1.10	< .001
Older ($\geq$ 65 years)	1.10	1.09–1.11	< .001

OR per 1 J/min increase in TWA mechanical power, adjusted for age, sex, predicted body weight, oxygenation ratio, vasopressor use, CRRT, and pulmonary disease phenotype. P-interaction: oxygenation 0.067; lung capacity 0.941; age 0.060. PF/SF, PaO_2/FiO_2 or SpO_2/FiO_2 ratio.

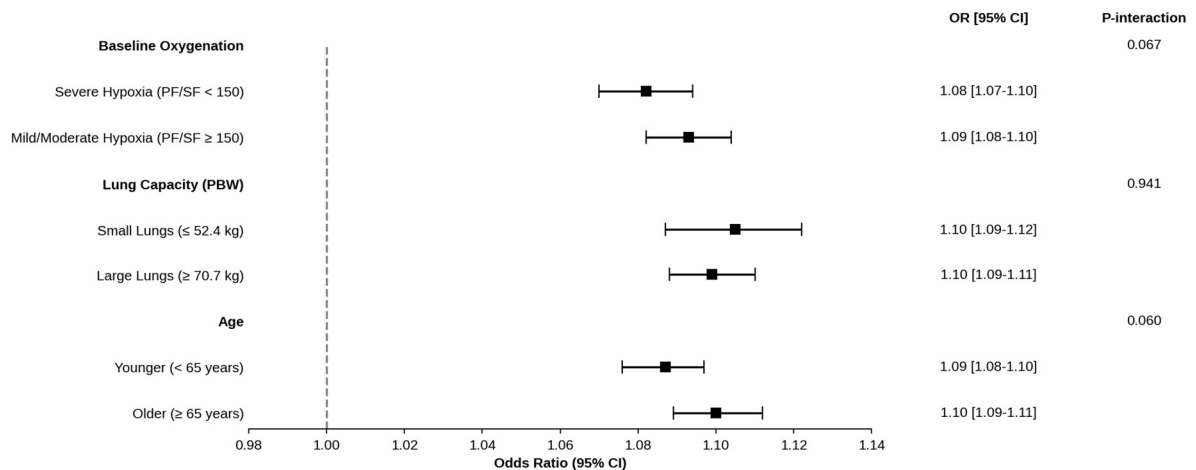


Figure 3. Forest plot of adjusted odds ratios for the association between time-weighted average mechanical power and 28-day mortality within pre-specified clinical subgroups. Squares denote point estimates and horizontal lines the 95% confidence intervals; the dashed line marks the null (OR = 1.00). P-interaction values are shown at right.

Discussion

Summary of Principal Findings

In this retrospective cohort of 18,980 mechanically ventilated adults, the mechanical power delivered during the first 24 hours of invasive ventilation was independently associated with 28-day all-cause mortality after adjustment for age, sex, predicted body weight, oxygenation, organ-support requirements, and pulmonary disease phenotype. Each 1 J/min increment in the time-weighted average (TWA) MP raised the adjusted odds of death by 9% (OR 1.09; 95% CI 1.09–1.10), and the association was reproduced almost exactly when the exposure was re-expressed as the daily median or as total cumulative energy, indicating that the finding is a property of the energy load itself rather than of any single way of summarizing it. The Kaplan–Meier analysis added two clinically important observations: the survival disadvantage was graded rather than threshold-like, and it was already evident across quartile boundaries that lay entirely below the 17 J/min value commonly cited as a danger threshold. Finally, the association was preserved, with point estimates compressed into a 1.08–1.10 band and all interaction tests non-significant, across strata of oxygenation severity, body size, and age, that is, across the principal axes along which clinical vulnerability is usually assumed to vary.

Comparison with Previous Studies

The central result aligns with, and materially extends, the foundational analysis of Serpa Neto and colleagues, who reported that mechanical power was independently associated with in-hospital mortality in 8,207 patients pooled from the MIMIC-III and eICU databases, with risk rising consistently beyond approximately 17 J/min [8]. The present estimate of a 9% increase in mortality odds per J/min is directionally concordant with that work; the larger and diagnostically unrestricted cohort analyzed here narrows the uncertainty around the effect and, through the quartile-based survival analysis, shows that the risk gradient is continuous and begins well below the 17 J/min mark rather than emerging only above it. This refinement is clinically consequential: it implies that a threshold framed at 17 J/min, as characterized in the multicenter prevalence survey of González-Castro and colleagues in which 35–41% of patients were ventilated above that

level [14], may understate the proportion of patients exposed to physiologically meaningful energy loads.

The cumulative-energy model is conceptually continuous with the time-varying analysis of Uner and colleagues, who demonstrated that each daily increment in mechanical power independently increased the hazard of death and that even brief exposure to higher-intensity ventilation was harmful [9]. Whereas that study integrated risk over the entire ventilation course, the present analysis isolates the first 24 hours and weights each setting by its true duration; the convergence of the two approaches strengthens the inference that it is accumulated mechanical energy, not a transient peak, that carries the mortality signal. The same logic is reinforced by the single-center work of Manrique and colleagues, in which the number of hours spent above 18 J/min predicted ICU mortality [13]; both their dose-by-time formulation and the duration-weighted TWA used here point to the same underlying construct of integrated exposure.

A more nuanced relationship exists with the report of Zhu and colleagues, who found that normalizing mechanical power to predicted body weight strengthened its association with mortality and argued that energy delivery should be scaled to lung size [10]. The present subgroup analysis does not support a strong version of that hypothesis: the adjusted odds ratio was identical (1.10) in the smallest and largest body-size quartiles, and the interaction test was the most clearly null of the three examined (p -interaction = 0.941). Within the body-size range represented in this cohort, the absolute rate of energy delivery, rather than its concentration per unit of anatomical lung volume, appears to be the operative driver of risk. This divergence is best read not as a contradiction but as a boundary condition on the normalization argument, which may matter most at the extremes of body habitus that a quartile contrast does not isolate.

The phenotype-independence of the association is consistent with several complementary lines of evidence. van Meenen and colleagues showed that mechanical power retained an independent relationship with mortality in invasively ventilated patients without ARDS [11], and Azizi and colleagues demonstrated that the MP–mortality association was not modified by a diagnosis of COVID-19 [12]. The non-significant ARDS coefficient in all three regression models here ($p = .471$ –.830), together

with the stable effect estimates across the pulmonary-phenotype adjustment, extends this pattern to a broad mixed-ICU population and supports the view that the injurious mechanism operates through delivered energy rather than through any single disease substrate. The apparently protective COPD coefficient in the present mixed-cohort model (OR 0.85) warrants careful interpretation, because it is a between-phenotype contrast (the relative position of COPD patients against the remainder of a heterogeneous cohort, conditional on mechanical power) and not a statement that energy delivery is benign in obstructive disease. The COPD-specific analysis of Wu and colleagues resolves this apparent paradox directly: in 1,704 invasively ventilated COPD patients drawn from the same MIMIC-IV source, mechanical power was independently associated with 28-day all-cause mortality (hazard ratio 1.04 per unit; 95% CI 1.03–1.05), the restricted cubic spline showed a linear, non-threshold dose-response (p for non-linearity = 0.967), and no subgroup interaction reached significance (p -interaction 0.114–0.967), with an optimal cut-off of 17.38 J/min [23]. Their within-disease findings mirror the present results in three respects: a linear rather than threshold-defined risk gradient, the absence of effect modification, and a danger zone close to 17 J/min, indicating that the protective phenotype coefficient observed here reflects baseline differences in case mix rather than any attenuation of the energy–mortality relationship within COPD itself.

Mechanistically, the epidemiological signal observed here is coherent with the experimental physiology that motivated the mechanical-power concept. Hotchkiss and colleagues showed in an isolated perfused-lung model that, at a fixed peak airway pressure, reducing ventilatory frequency markedly diminished edema and hemorrhage, establishing that the rate at which energy is applied is an independent determinant of injury [2]. Mechanical power, by construction, incorporates respiratory rate and flow alongside volume and pressure, and the unifying-variable hypothesis advanced by Gattinoni and colleagues holds that these components are most informative when considered together rather than in isolation [6]. The present finding, that mortality tracks an energy-rate parameter even when its individual components may individually remain within conventionally protective limits, provides large-scale clinical support for that hypothesis and complements the established literature on the static determinants of injury, namely the survival benefit of lower tidal volume demonstrated by the ARDS

Network [3] and the dominant prognostic role of driving pressure identified by Amato and colleagues [4]. Read alongside the LUNG SAFE finding that mortality remains high and lung-protective targets are inconsistently achieved in routine practice [5], these data offer a partial explanation for residual mortality under contemporary ventilation: harmful energy delivery persists even where individual parameter ceilings are nominally respected.

Clinical Implications

The most directly actionable implication is that the prognostic relevance of mechanical power is not confined to any identifiable high-risk subgroup. Because the association held without meaningful attenuation across oxygenation severity, body size, and age, monitoring of delivered energy should not be reserved for patients with severe ARDS or a priori risk factors but considered across the ventilated population. The practicality of this is supported by the validation work of Chiumello and colleagues, who showed that the surrogate formula approximates the geometric reference method closely enough for routine use in both volume- and pressure-controlled ventilation [7], and by the fact that the equation requires only parameters already available at the bedside [6]. The time-weighted average is particularly suited to this role: by weighting each setting by its true duration, it represents the energy actually experienced by the respiratory system over the first day more faithfully than a single spot value, and it expresses risk on the J/min scale at which clinicians already reason about ventilator settings. Incorporating TWA MP as a complementary surveillance parameter alongside tidal volume, plateau pressure, and driving pressure may identify patients at elevated risk even when each individual limit is respected, a group that the prevalence data of González-Castro and colleagues suggest is substantial [14].

Strengths

Several features strengthen the inferences drawn here. The cohort is large and diagnostically heterogeneous, comprising the full spectrum of ICU pathology rather than a single disease, which improves external validity relative to phenotype-restricted studies. Invasive ventilation was anchored to physical endotracheal-intubation timestamps rather than to charting flags alone, reducing exposure misclassification. The exposure was

computed as a duration-weighted average over the first 24 hours, correcting for the change-triggered charting intervals of the source database, a methodological refinement over analyses that apply a simple arithmetic mean to rows representing unequal exposure times. Three independently constructed exposure metrics were pre-specified and analyzed in parallel, and their concordance guards against the result being an artifact of a single operationalization. Finally, the subgroup framework was pre-specified, applied with consistent covariate adjustment, and accompanied by formal interaction testing, allowing effect modification to be assessed without the inflation that finer post hoc stratification would introduce.

Limitations

This study has several limitations that bear on interpretation. First, the underlying data derive from a single tertiary center, and ventilator-management culture, case mix, and outcomes may differ elsewhere; external validation in multicenter cohorts is required before generalization. Second, mechanical power was calculated from a dynamic surrogate equation based on peak rather than plateau pressure; in patients with elevated airway resistance this will incorporate a resistive component and may overestimate the energy transmitted to the alveolar compartment, although the same formulation has been validated for clinical use across ventilation modes [6,7]. Third, as an observational analysis it remains susceptible to residual and unmeasured confounding, particularly confounding by indication: sicker patients may receive both more aggressive ventilation and worse outcomes for reasons not fully captured by the recorded covariates, and sedation depth and clinician ventilation strategy were not available. Fourth, the analysis does not capture spontaneous breathing effort or patient–ventilator asynchrony, both of which contribute to the true energy load and would, if anything, be unmeasured sources of additional power. Fifth, severity of illness was represented by vasopressor and CRRT proxies because the Sequential Organ Failure Assessment score was too incomplete to model without substantial sample loss; residual severity confounding therefore cannot be excluded. Sixth, the oxygenation covariate combined arterial $\text{PaO}_2/\text{FiO}_2$ with imputed $\text{SpO}_2/\text{FiO}_2$ values, introducing measurement heterogeneity even though the substitution itself is validated [21]. Seventh, pulmonary phenotypes were derived from administrative ICD codes and are subject to misclassification; the apparently protective associations for

COPD, asthma, and heart failure most plausibly reflect between-phenotype differences in baseline case mix and confounding by indication rather than genuinely protective biology; as the COPD-specific analysis of Wu and colleagues demonstrates [23], should not be read as evidence that mechanical power is harmless within any of these diagnostic groups. Eighth, the requirement for at least four hours of valid mechanical-power data excludes very brief ventilation courses and may select against the most rapidly extubated or most rapidly dying patients. Ninth, the exposure window was restricted to the first 24 hours; whether mechanical-power trajectories on subsequent days carry additional independent prognostic information was not examined and remains an open question.

Conclusions

This retrospective analysis of 18,980 mechanically ventilated adults answers the two questions it set out to address. First, the mechanical power delivered during the first 24 hours of invasive ventilation was independently associated with 28-day all-cause mortality: each 1 J/min increment in the time-weighted average raised the adjusted odds of death by 9% (OR 1.09; 95% CI 1.09–1.10), and the relationship was reproduced without material change when the exposure was expressed as the daily median or as total cumulative energy, confirming that the signal is a property of the energy load rather than of any single way of summarizing it. Second, the association was not modified by the clinical characteristics along which susceptibility is usually presumed to differ (oxygenation severity, body size, and age) with effect estimates compressed into a narrow 1.08–1.10 band and every interaction test non-significant. Taken with the graded survival disadvantage evident on Kaplan–Meier analysis even across power values below the conventionally cited 17 J/min threshold, these findings indicate that no readily identifiable subgroup of ventilated patients is exempt from the mortality risk associated with excess delivered energy, and that the first day of ventilation is an actionable window for energy-aware lung protection.

These results support the integration of time-weighted mechanical power monitoring into routine ventilator surveillance, as a complement to — not a replacement for — established tidal-volume, plateau-pressure, and driving-pressure targets. Several priorities for future work follow directly from the limitations of this study. External

validation in multicenter, demographically distinct cohorts is required before the magnitude of the association can be generalized. Prospective interventional trials are needed to determine whether actively titrating ventilation to reduce mechanical power changes outcomes, since an observational association cannot establish that lowering power is itself protective. The prognostic contribution of mechanical-power trajectories beyond the first 24 hours, and of the energy added by spontaneous breathing effort and patient–ventilator asynchrony, should be quantified, as neither was captured here. Finally, the demonstration that mortality risk accrues below 17 J/min indicates that empirically derived, phenotype-specific safety thresholds, rather than a single universal cut-off, warrant formal investigation.

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